

①

$$P_{\text{diss}} = 2 \text{ mW} \quad V_{\text{DD}} = 5 \quad \left[I_{\text{D}} = 0.8 \text{ mA} \right]$$

$$V_{\text{OL}} = 1 \text{ V}$$

$$I_{\text{D}} (V_{\text{in}} = V_{\text{DD}}) = \frac{K_{\text{ad}}}{2} [V_{\text{gs}} - |V_{\text{tp}}|]^2$$

$$\left(\frac{W}{L} \right)_p = 10 \quad \text{--- (4)}$$

$$V_{\text{OL}} = 1 \text{ (given)}$$

$$\frac{2.5}{2} \left(\frac{W}{L} \right)_n \left[2(V_{\text{DD}} - V_{\text{tn}}) V_{\text{OL}} - V_{\text{OL}}^2 \right] = \frac{10}{2} \times 10 \left[5 - 1 \right]^2$$

$$\left(\frac{W}{L} \right)_n = 9.7 \rightarrow \text{(4) if } L = 1 \mu\text{m}$$

$$(W)_p = 10 \mu\text{m}$$

$$(W)_n = 9.7 \mu\text{m}$$

fall time

$$(b) I_{\text{avg}} = \frac{1}{2} \left[I(V_{\text{in}} = 5, V_{\text{out}} = 0.9 V_{\text{DD}}) + I(V_{\text{in}} = 5, V_{\text{out}} = 0.1 V_{\text{DD}}) \right]$$

$$I_{\text{avg}} = 4.365 \text{ KHz}$$

$$\tau_{\text{fall}} = \frac{C \Delta V}{I_{\text{avg}}}$$

$$\tau_{\text{fall}} = \frac{0.549 \text{ C}}{\text{KHz}} \quad \text{(4)}$$

rise time

$$I_{\text{avg}} = \frac{1}{2} \left[I(V_{\text{in}} = 1, V_{\text{out}} = 0.3) + I(V_{\text{in}} = 1, V_{\text{out}} = 2.7) \right]$$

$$I_{\text{avg}} = 0.377 \text{ KP}$$

$$\tau_{\text{rise}} = \frac{6.36 \text{ C}}{\text{KP}} \quad \text{(4)}$$

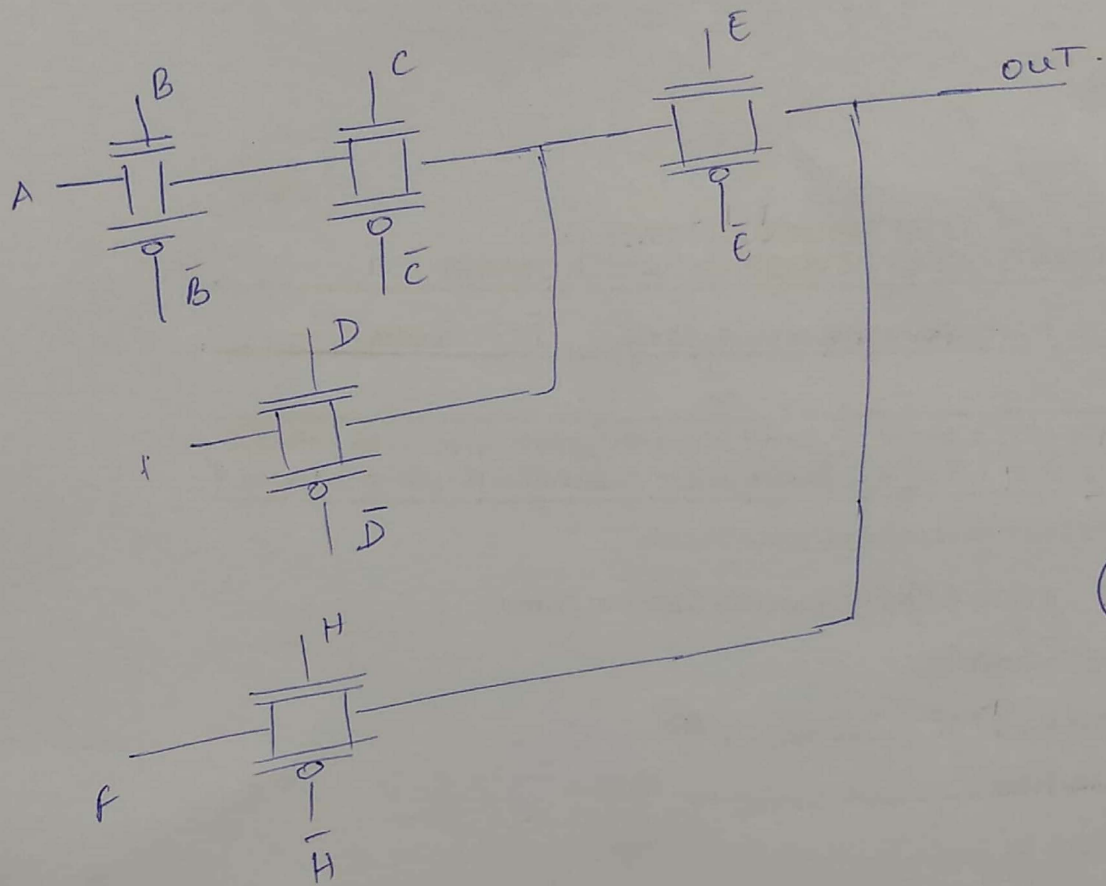
$$\tau_{\text{rise}} = \tau_{\text{fall}}$$

$$\frac{0.549 \text{ K}}{\text{KHz}} = \frac{6.36 \text{ K}}{\text{KP}}$$

$$\frac{\text{KP}}{\text{KHz}} = 11.58$$

$$\frac{W_p}{W_n} = 28.95 \quad \text{(2)}$$

Q.2

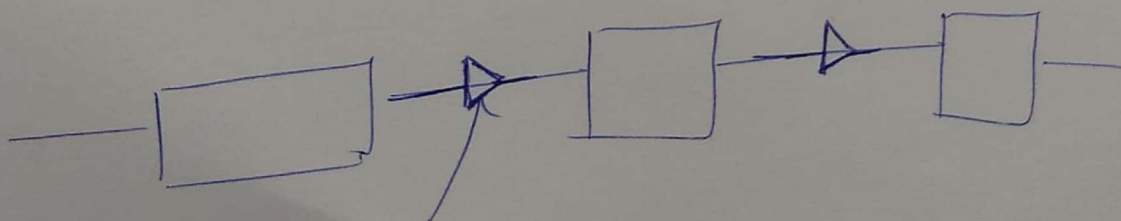


(4)

$$M_{opt} = 1.7 \sqrt{\frac{t_{buff}}{R \cdot C}}$$

$$R = R_{eq,n} \parallel R_{eq,p} = 9 \text{ k}\Omega$$

$$M_{opt} = 3.08 \quad (2)$$



insert buffers.

(1)

Q3

$$V_{DD} = 2.5V, V_{th} = 0.4V, \mu_n C_{ox} = 115 \mu A/V^2,$$

$$P = 4 mW \Rightarrow I_D V_{DD} = 4 mW \Rightarrow I_D = 1.6 mA$$

a) For sat. $V_{DS} \geq V_{GS} - V_{th}$

$$V_{in,max} \text{ for Sat.} = 1.8 V$$

$$\Rightarrow 2.5 - I_D R_L = V_{GS} - V_{th} = 1.8 - 0.4$$

$$\Rightarrow I_D R_L = 1.1 \Rightarrow R_L = \frac{1.1 V}{1.6 mA} = \underline{\underline{687.5 \Omega}}$$

b) $I_D = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L}\right) (V_{GS} - V_{th})^2$

$$\Rightarrow \frac{W}{L} = \frac{2 \times 1.6 \times 10^{-3}}{115 \times 10^{-6} \times 1.4^2} = 14.2 \approx \underline{\underline{14}}$$

c)

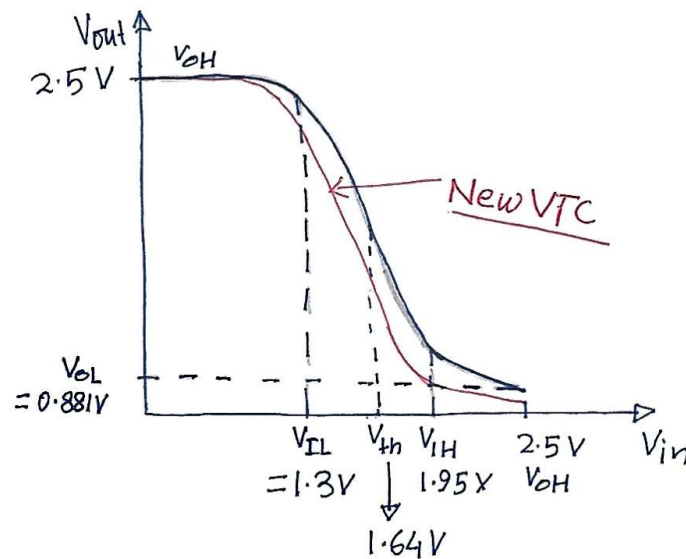
$$V_{OH} = V_{DD} = 2.5 \text{ V} \quad | \quad \frac{1}{k_n R_L} = 0.903 \text{ V}$$

$$V_{IL} = V_{t_{on}} + \frac{1}{k_n R_L} = 0.4 + 0.903 = \underline{\underline{1.303 \text{ V}}}$$

$$V_{IH} = V_{t_{on}} + \sqrt{\frac{8}{3} \frac{V_{DD}}{k_n R_L}} - \frac{1}{k_n R_L} = \underline{\underline{1.95 \text{ V}}}$$

$$V_{OL} = V_{DD} - V_{t_{on}} + \frac{1}{k_n R_L} + \sqrt{\left(V_{DD} - V_{t_{on}} + \frac{1}{k_n R_L}\right)^2 - \frac{2V_{DD}}{k_n R_L}} = \underline{\underline{0.881 \text{ V}}}$$

$$V_m \text{ or } V_{th} = V_{t_{on}} - \frac{1}{k_n R_L} + \sqrt{\left(V_{t_{on}} - \frac{1}{k_n R_L}\right)^2 + \frac{2V_{DD}}{k_n R_L}} - V_{t_{on}}^2 = \underline{\underline{1.644 \text{ V}}}$$



d) $\frac{W}{L}$ increased by 20% $\Rightarrow \frac{1}{k_n R_L} = 0.7525 \text{ V}$

$\Rightarrow$ New $V_m = 1.58 \text{ V}$

$V_{IL} \downarrow, V_{OL} \downarrow, V_{IH} \downarrow$

a

$$C_{db} = AC_{j0} K_{eq} + PC_{jsw} K_{eq} (sw)$$

$$A = 60 \mu m^2$$

$$P = 6 + 6 + 10 = 22 \mu m$$

$$C_{db} = 60 \times 0.95 \times 0.44 + 22 \times 3.5 \times 0.46$$

$$= 25.08 + 35.42$$

$$C_{db} = 60.50 pF$$

(b)

Scaling factor = 10.

$$\text{SiO}_2 \text{ thickness of scaled device} = \frac{0.05 \mu\text{m}}{10} = 5 \text{ nm}.$$

$$\frac{\epsilon_{\text{SiO}_2}}{t_{\text{SiO}_2}} = \frac{\epsilon_{\text{ZnO}_2}}{t_{\text{ZnO}_2}}$$

$$t_{\text{ZnO}_2} = \frac{\epsilon_{\text{ZnO}_2} \times t_{\text{SiO}_2}}{\epsilon_{\text{SiO}_2}}$$

$$= \frac{25}{3.9} \times 5 = 32.05 \text{ nm}.$$

Now, Electric field in ZnO_2 is -

$$\frac{V}{d} = \frac{5}{32.05 \times 10^{-7} \text{ cm}} = 1.56 \times 10^6 \text{ V/cm}$$

As we know that for typical dry oxidation process

$$\frac{t_{Si}}{t_{SiO_2}} = 0.45$$

It is given that $t_{SiO_2} = 0.05 \mu m = 50 nm$

$$t_{Si} = 0.45 \times 50 = 22.5 nm = 22.5 \times 10^{-7} cm$$

$$\& A = 40 \mu m^2 = 40 \times 10^{-8} cm^2 = 4 \times 10^{-7} cm^2$$

$$\Rightarrow \text{Volume} = 22.5 \times 10^{-7} \times 4 \times 10^{-7} cm^3 \\ = 90 \times 10^{-14} = 9 \times 10^{-13} cm^3$$

$$\text{Mass} = 9 \times 10^{-13} \times 2.33 \\ = 2.097 \times 10^{-12} g$$

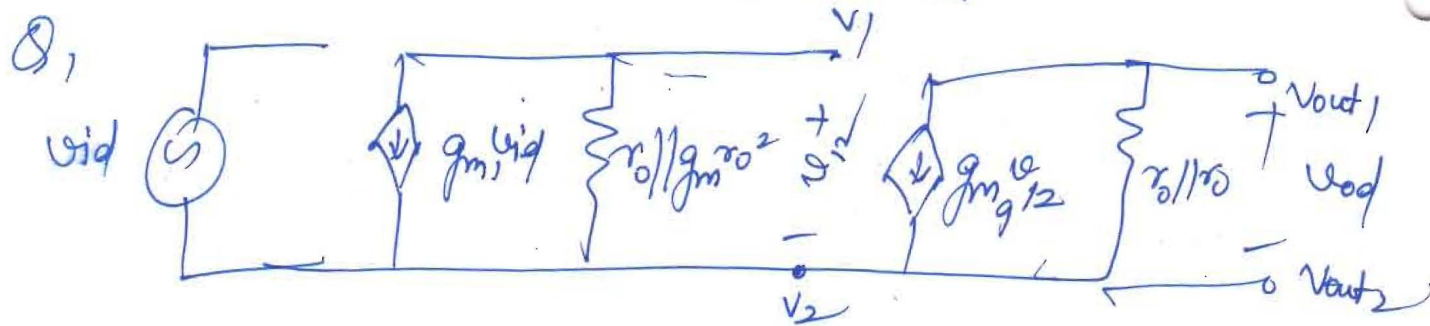
$$\text{Mole} = \frac{2.097 \times 10^{-12}}{28} = 0.0748 \times 10^{-12} \text{ mole}$$

$$Si \text{ Atoms} = \text{Mole} \times N_A$$

$$= 0.0748 \times 10^{-12} \times 6.023 \times 10^{23}$$

$$= 0.451 \times 10^{11} = 4.51 \times 10^{10} \text{ atoms}$$

PART B Sol



$$A_v(\text{max.}) = 5000$$

$$\text{Max input (for undistorted o/p)} = \frac{V_{out \text{ swing}}}{A_v} = 4.6 \text{ mV p-p}$$

$$\omega_{out} \Rightarrow 0.2 \text{ M rad/sec} = \omega_{-3dB}$$

$$\omega_{1,2} = 1 \text{ M rad/sec}$$

$$\omega_{GB} = 1 \text{ G rad/sec} = A_v \omega_{-3dB}$$

Shift ω_{GB} to 1 M rad/sec

$$\text{New } \omega_{-3dB} (\text{reqd.}) = 0.2 \times 10^3 \text{ rad/sec}$$

$$\text{So } C_{\text{additional}} = 99.9 \text{ nF}$$

$$C_i = C_L$$

Q.1.

$$Z_{out} = (S_1 \text{ open}) = \frac{1}{sC_L}$$

$$Z_{out} (S_1 \text{ closed}) = \frac{1}{sC_L} \parallel (R_1 + \frac{1}{sC_1})$$

$$= \frac{1}{s(2C_L)} \frac{(1 + sR_1C_L)}{(1 + s\frac{R_1C_1}{2})}$$

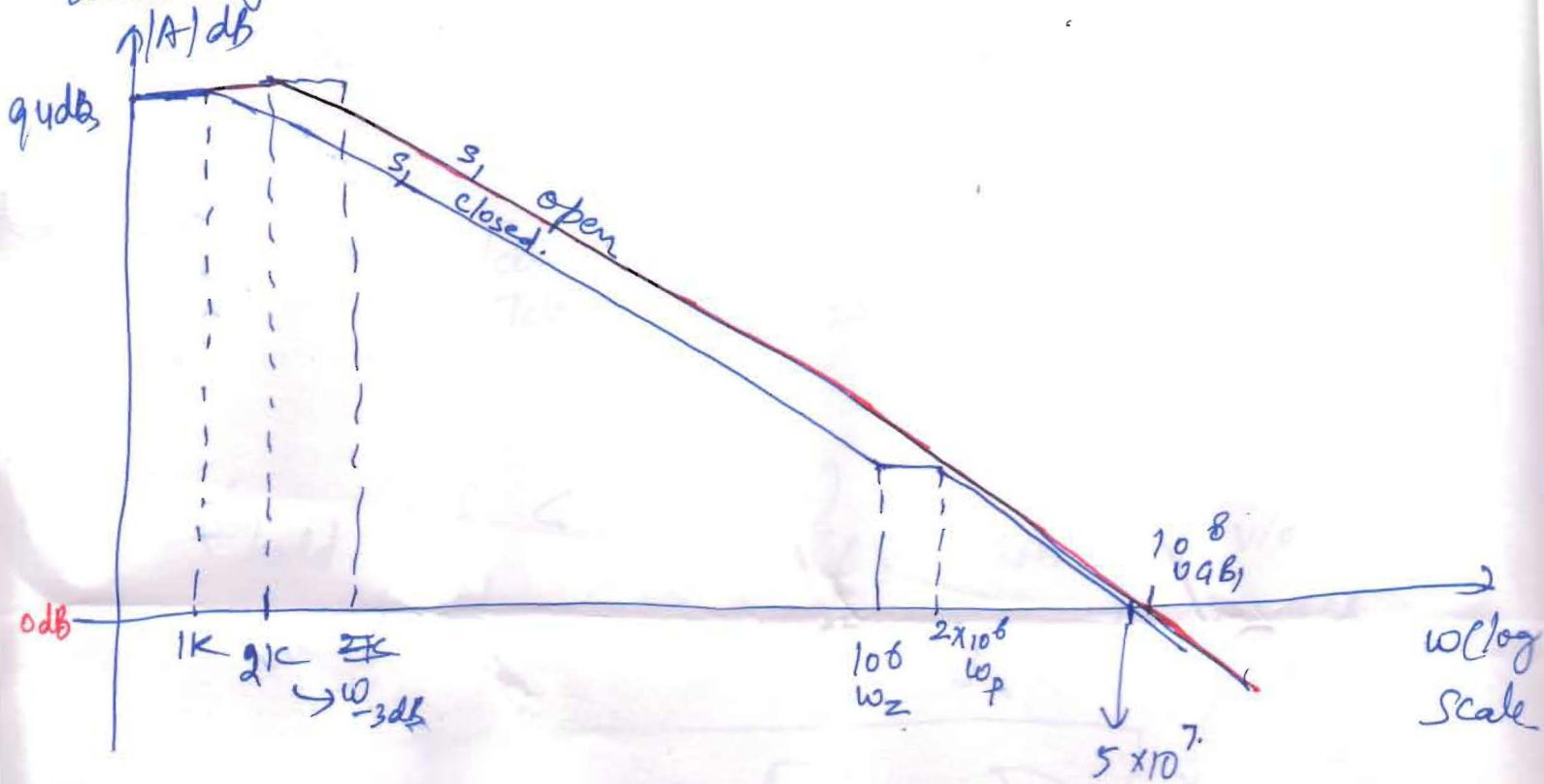
$$A_0 = 50,000 = (94 \text{ dB})$$

$$\omega_{out}(S_1 \text{ open}) = 2 \text{ K}, \quad \omega_{out}(S_1 \text{ closed}) = 1 \text{ K}$$

$$\omega_z = (1 + s R_1 C_1) = 10^6 \text{ rad/sec}$$

$$\omega_p = (1 + s \frac{R_1 C_L}{2}) = 2 \times 10^6 \text{ rad/sec}$$

Bode Mag. plot (corner)



Q.2

Q.
(a)

Path

A → = 44 ns delay max.

P → = 43 ns (max)

(C →)₁ = 45 ns (max)

(C →)₂ = 22 ns (max)

$t_c = 10 \text{ ns}$

(b) T_{clk} required

$$T_{clk} + \delta \geq T_{C-Q} + t_{logic} + t_{su}$$

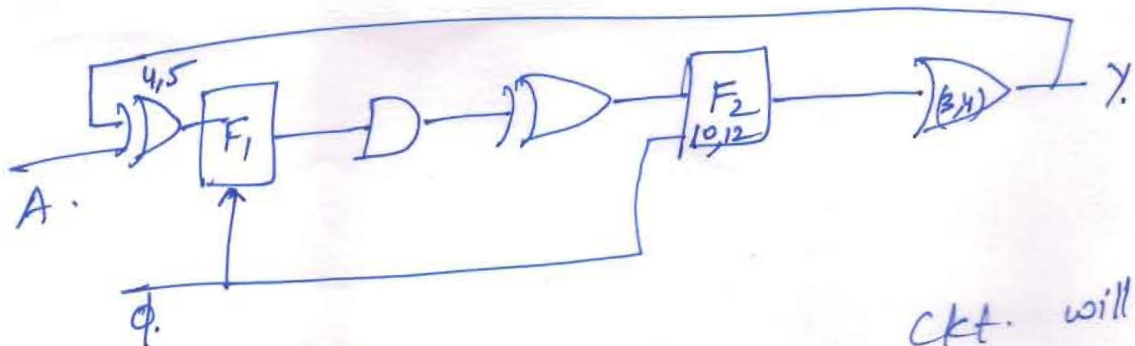
$$T_{clk} + 9 \text{ ns} \geq (9 + 14 + 8) \text{ ns}$$

$$T_{clk} \geq 22 \text{ ns} \Rightarrow f \geq 45 \text{ MHz}$$

(c) $t_{hold} + \delta < 15 \text{ ns}$ (min)

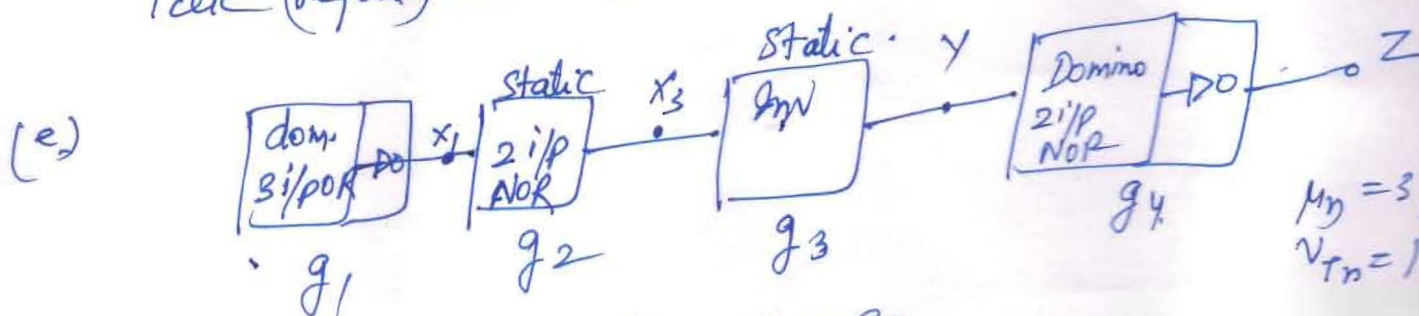
$t_{hold} < 6 \text{ ns}$

(Yes hold time violation)
given $t_{hold} = 8 \text{ ns}$



Ckt. will not work

T_{clk} (required) now $\geq 29 \text{ ns}$



$$\text{Total } G = g_1 \times g_2 \times g_3 \times g_4$$

$$= \frac{2}{4} \times \frac{7}{4} \times 1 \times \frac{2}{4} = \frac{28}{64}$$

$M_n = 3$
 $V_{Tn} = 1$